

AI Face Recognition Smart Door Lock

Project Objective

To design and build a low-cost, smart access control system that uses facial recognition to automatically unlock a door.

Key Features

- **Facial Recognition:** A small camera recognizes approved faces and automatically triggers the door to open.
- **Browser Dashboard:** A live-streaming web interface to monitor activity and manage users in real time.
- **Easy User Setup:** Add new people by taking a few quick snapshots, or remove access instantly with a single click.
- **Remembers Who You Add:** Saves your user list permanently so you won't lose your settings if the device loses power.
- **Compact PCB-Integrated Design:** A custom circuit board houses all components in a sleek, space-saving enclosure that fits discreetly near your door.

Core Components

Component	Function
ESP32-CAM Module (OV2640)	Wi-Fi MCU with camera for image capture and on-board AI face recognition
12V Electronic Door Lock	Solenoid-driven strike that physically secures/releases the door
TIP122 (NPN Darlington Transistor)	Switches the 12V lock coil from a GPIO-level signal
1N4007 Flyback Diode	Suppresses inductive kickback from the lock coil to protect TIP122
7805 5V Voltage Regulator	Steps 12V DC down to a regulated 5V rail for the ESP32-CAM
220uF Electrolytic Capacitor	Filters/smooths the regulated 5V output for stable operation
10k Resistor	Base resistor limiting drive current into the TIP122
220 Ohm Resistors (x2)	Current-limiting resistors for the red and green status LEDs
Red & Green LEDs	Visual access status indicators (denied / granted)
12V DC Power Input	Main external supply powering the lock and the ESP32-CAM

Block Diagram

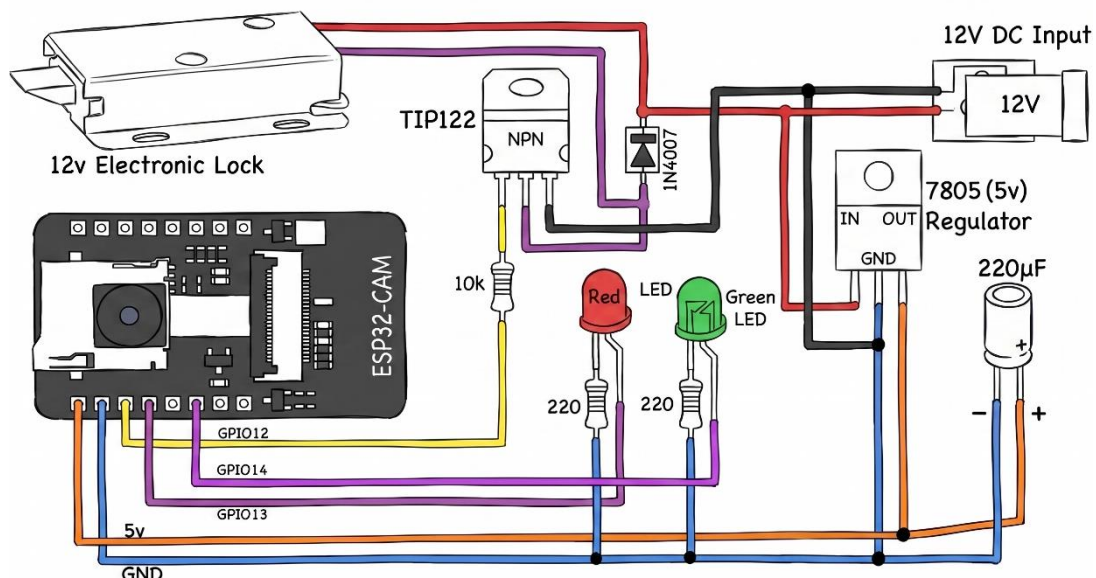


Figure 1- Block Diagram (AI Generated)